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Handheld strapping device

Abstract

Strapping devices and methods are provided. The strapping device can include a member. The strapping device can include a first actuator coupled with the member to move the member between a first position and a second position. The strapping device can include a sensor to detect an electrical characteristic of the first actuator and to transmit a signal indicative of the electrical characteristic. The strapping device can include a data processing system. The data processing system can receive the signal and determine the member is in contact with a welding component based on the electrical characteristic. The data processing system can determine the second position of the member based on the member being in contact with the welding component. The data processing system can transmit a control signal to the first actuator and a second actuator to initiate a welding cycle with the member at the second position.

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Background/Summary

BACKGROUND

(1) Devices can be used to secure straps around objects. The straps can be secured to one another.

SUMMARY

(2) At least one aspect is directed to a strapping device. The strapping device can include a grip member. The strapping device can include a first actuator coupled with the grip member to move the grip member between a first position and a second position. The strapping device can include a sensor to detect an electrical characteristic of the first actuator and transmit a signal indicative of the electrical characteristic. The strapping device can include a data processing system communicably coupled with the first actuator and the sensor. The data processing system can receive the signal indicative of the electrical characteristic. The data processing system can determine the grip member is in contact with a welding component based, at least partially, on the electrical characteristic. The data processing system can transmit a first control signal to the first actuator to position the grip member at the second position based on the grip member being in contact with the welding component. The data processing system can transmit a second control signal to the first actuator and a second actuator to initiate a welding cycle to weld the welding component such that the welding cycle initiates with the grip member at the second position.

(3) At least one aspect is directed to a method. The method can include moving a grip member of a strapping device from a first position toward a welding component via an actuator of the strapping device. The method can include determining the grip member is contacting the welding component. The method can include moving the grip member to a second position based on the grip member contacting the welding component. The method can include initiating a welding cycle to friction weld the welding component such that the welding cycle initiates with the grip member at the second position.

(4) At least one aspect is directed to a welding system. The welding system can include a data processing system. The data processing system can be communicably coupled with a sensor, a first actuator, and a second actuator. The sensor can detect an electrical characteristic of the first actuator indicative of a grip member of a welding assembly contacting a welding component. The first actuator can move the grip member. The second actuator can move a portion of the welding component.

The data processing system can receive a signal indicative of the electrical characteristic. The data processing system can determine the grip member is in contact with the welding component based, at least partially, on the electrical characteristic. The data processing system can transmit a first control signal to the first actuator to position the grip member at a second position based on the grip member being in contact with the welding component. The data processing system can transmit a second control signal to the first actuator and the second actuator to initiate a welding cycle to weld the welding component such that the welding cycle initiates with the grip member at the second position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** depicts an example strapping device, according to some aspects.
(2) FIG. **2**A is a schematic diagram of an example strapping device, according to some aspects.
(3) FIG. **2**B is a schematic diagram of an example strapping device, according to some aspects.
(4) FIG. **3** depicts side views of example positions of a grip member, according to some aspects.
(5) FIG. **4** is a schematic diagram of an example data processing system, according to some aspects.
(6) FIG. **5** is a flow diagram depicting an example method, according to some aspects.
(7) FIG. **6** is a flow diagram depicting an example method, according to some aspects.
DETAILED DESCRIPTION
(8) Following below are more detailed descriptions of various concepts related to, and implementations of, methods, apparatuses, and systems of welding a component via friction welding. The various concepts introduced above and discussed in greater details below may be implemented in any of numerous ways.
(9) The present disclosure is directed to systems and methods for using a computing system to detect a thickness of a welding component (e.g., a strap) that is to be friction welded and automatically adjusting a strapping device to accommodate the thickness of the welding component to achieve a proper weld while reducing the amount of additional wear experienced by the strapping device. The disclosed solutions can have the technical advantage of modifying mechanical aspects of a strapping device without manually modifying physical parts of the strapping device (e.g., adding/removing parts). The disclosed solutions can perform a custom weld cycle with custom welding parameters that are specific to the welding component and the strapping device without having to manually adjust the strapping device. As such, a single strapping device can automatically modify mechanical aspects and welding parameters to create a proper weld for welding components with various thicknesses. For example, the disclosed solutions can prevent the strapping device from starting a welding cycle too soon such that the welding cycle results in scuffing and under-welding and prevent the strapping device from starting a welding cycle too late such that there is too much friction in the device such that the motor cannot overcome the friction to initiate the welding cycle.
(10) The disclosed solution can include a strapping device. The strapping device can be or include a friction welding strapping device. The strapping device can include a weld assembly. The weld assembly can include a first actuator (e.g., a servo motor), a cam to be driven by the first actuator, and a member to be driven by the cam. The member can move within the strapping device to contact a welding component (e.g., a component to be welded). The member can start in a first position (e.g., a retracted/open/stored position) such that the welding component can be positioned within the strapping device. The member can move in a first direction (e.g., toward the welding component) to contact the welding component. The strapping device can include a data processing system. The data processing system can detect when the member contacts the welding component. For example, the strapping device can include a sensor that detects an electrical characteristic (e.g., a current) of the first actuator. The sensor can transmit a signal indicative of the electrical characteristic to the data processing system. The data processing system can compare the electrical characteristics to a threshold indicative of the first actuator contacting the welding component. For example, the threshold can be a current value or a change in current. The threshold can be tailored to the first actuator or the strapping device (e.g., based on age of actuator or number of welds performed by the actuator). When the electrical characteristic meets or exceeds the threshold, the data processing system can determine that the arm is in contact with the welding component.
(11) The data processing system can determine a thickness of the welding component. For example, the device can have a base. The welding component can be disposed between the base and the member. The data processing system can determine a distance between the base and the member when the member contacts the welding component. For example, the data processing system can determine how far the member has traveled from the first position when the member contacts the welding component. The distance can be the thickness of the welding component. Based on the thickness of the welding component, the data processing system can determine a second position for the member. The second position can be a motor starting position for the member to facilitate proper friction welding of the welding component. For example, the second position can prevent the member from applying too little pressure to the welding component to prevent a final weld from being insufficient and can prevent the member from applying too much pressure to the welding component such that the member does not prevent the welding component from moving to create the friction needed to create the weld.
(12) The strapping device can include a second actuator. With the member in the second position, the data processing system can transmit a control signal to the first actuator and the second actuator to initiate a welding cycle. The second actuator can move a first portion of the welding component to create friction between the first portion and a second portion. The first actuator can move the member toward the welding component to a third position to increase the friction between the first portion and the second portion. The member can remain in the third position during a cooling portion of the welding cycle.
(13) The data processing system can perform real-time signal processing to determine the thickness of the welding component, adjust the strapping device based on the thickness of the welding component, and perform a customized welding cycle to

generate a proper friction weld. These determinations and adjustments can be specifically tailored for any individual strapping device.

(14) FIG. **1** depicts an example device, shown as strapping device **100**. The strapping device **100** can be handheld. For example, the strapping device **100** can be less than a threshold weight to facilitate manual manipulation of the strapping device **100** with a single hand. The strapping device **100** can have a body **105**. The body **105** can be a housing for the various mechanisms and components described herein to facilitate a welding cycle to weld a welding component **110**. For example, the body **105** can include a base member **115**. The body **105** can receive a welding component **110** to be welded and the base member **115** can support the welding component **110**. The welding component **110** can include a first portion, shown as base portion **120**, and a second portion, shown as actuator portion **125**, to be welded together. The strapping device **100** can include a grip member **130**. The grip member **130** can selectively engage and disengage with the welding component **110** to facilitate the welding cycle.

(15) The strapping device **100** can include at least one actuator. For example, the strapping device **100** can include a first actuator, shown as member actuator **135**. The member actuator **135** can move the grip member **130** to selectively engage and disengage the welding component **110** to provide a desired amount of friction between the base portion **120** and the actuator portion **125** of the welding component **110**. The strapping device **100** can include a second actuator, shown as weld actuator **140**. The weld actuator **140** can selectively engage and disengage with a portion of the welding component **110** (e.g., the actuator portion **125**) to create friction between the base portion **120** and the actuator portion **125** to generate a friction weld.

(16) The strapping device **100** can include at least one sensor **145**. The sensor **145** can detect a characteristic of a component of the strapping device **100**. For example, the sensor **145** can detect an electrical characteristic of the member actuator **135** (e.g., current drawn by the member actuator **135**). The strapping device **100** can include at least one data processing system **150**. The data processing system **150** can receive, store, and analyze data associated with the strapping device **100** and the welding component **110** to determine at least one welding cycle parameter. For example, the welding cycle parameter can include a motor starting position, a welding time, a cooling time, a holding position, a welding pressure, and an actuator speed, among others. The data processing system **150** can generate and transmit at least one control signal to actuate at least one component of the strapping device **100** according to the welding cycle parameter.

(17) FIG. **2**A depicts an example schematic diagram of the strapping device **100**. The body **105** can receive the base portion **120** and the actuator portion **125** of the welding component **110**. For example, the base portion **120** can be a first end or a first portion of a strap and the actuator portion **125** can be a second end or a second portion of the same strap or the base portion **120** can be an end or a portion of a first strap and the actuator portion **125** can be an end or a portion of a second, different strap. The strapping device **100** can weld the portions **120**, **125** of the welding component **110** together. For example, the strapping device **100** can weld the base portion **120** with the actuator portion **125** via friction welding.

(18) The base member **115** can interface with the base portion **120** of the welding component **110** (e.g., first end of a strap or end of a first strap). The base portion **120** of the welding component **110** can be disposed between the base member **115** and the actuator portion **125** of the welding component **110** (e.g., second end of the same strap or end of a second strap). The base member **115** can support the base portion **120** and the actuator portion **125** of the welding component **110**.

(19) The base member **115** can include a grip, shown as base grip **205**. The base grip **205** can be coupled with or integral with the base member **115**. The base grip **205** can interface with the base portion **120** of the welding component **110**. The base grip **205** can facilitate holding the base portion **120** in place during a welding cycle. The base grip **205** can be any material or texture that can increase a friction between the base member **115** and the base portion **120** of the welding component **110** to prevent the base portion **120** from moving relative to the base member **115** during a welding cycle.

(20) The grip member **130** of the strapping device can interface with an opposite side of the welding component **110** than the base member **115**. For example, the grip member **130** can interface with the actuator portion **125** of the welding component **110**. For example, a first end of a strap can interface with the base member **115** and a second end of the strap can interface with the grip member **130**. The base portion **120** and the actuator portion **125** of the welding component **110** can be disposed between the base member **115** and the grip member **130**.

(21) The grip member **130** can include or be coupled with a compressible member **210**. The compressible member **210** can be disposed between the grip member **130** and the member actuator **135**. The member actuator **135** can be coupled with the grip member **130** via the compressible member **210**. The compressible member **210** can be, for example, a spring. The member actuator **135** can compress or decompress the compressible member **210** to adjust a pressure (e.g., the welding pressure) that the grip member **130** applies to the welding component **110**. For example, member actuator **135** can move the grip member **130** such that the grip member **130** can contact the welding component **110**. With the grip member **130** in contact with the welding component **110**, the member actuator **135** can compress the compressible member **210** to increase a pressure applied to the welding component **110** via the grip member **130** or decompress the compressible member **210** to decrease the pressure applied to the welding component **110**. The grip member **130** can remain stationary as the member actuator **135** adjusts the compressible member **210**.

(22) The grip member **130** can include a grip, shown as member grip **215**. The member grip **215** can be coupled with or integral with the grip member **130**. The member grip **215** can interface with the actuator portion **125** of the welding component **110**. The member grip **215** can increase friction between the actuator portion **125** and the grip member **130** during the welding cycle. The base portion **120** and the actuator portion **125** can be disposed between the base grip **205** and the member grip **215**. The member grip **215** can be any material or texture that can increase a friction between the grip member **130** and the actuator portion **125** of the welding component **110**.

(23) The member actuator **135** of the strapping device **100** can be coupled with the grip member **130** to move the grip member **130** within the body **105**. For example, the member actuator **135** can move the grip member **130** between a first position and a second position to selectively engage and disengage the welding component **110**. The member actuator **135** can be coupled

with the compressible member **210** of the grip member **130**. For example, the member actuator **135** can move the grip member **130** via the compressible member **210**. With the grip member **130** in contact with the welding component **110**, the member actuator **135** can continue to move (e.g., compress or decompress) the compressible member **210** to adjust the pressure applied to the welding component **110**.

(24) The strapping device **100** can include a cam **220**. The member actuator **135** can be coupled with the grip member **130** via the cam **220**. For example, the cam **220** can be disposed between the member actuator **135** and the grip member **130**. The cam **220** can be disposed between the member actuator **135** and the compressible member **210**. The member actuator **135** can be coupled with the cam **220** and can rotate the cam **220** such that the cam **220** can move (e.g., push or pull) the grip member **130** in a desired direction. With the grip member **130** in contact with the welding component **110**, the cam **220** can continue to move (e.g., compress or decompress) the compressible member **210** to adjust the pressure applied to the welding component **110**.

(25) The weld actuator **140** of the strapping device **100** can couple with or interface with at least one of the welding component **110** and the base member **115** to facilitate welding of the welding component **110**. For example, the weld actuator **140** can engage the actuator portion **125** of the welding component **110** to move the actuator portion **125** relative to the base portion **120** to create friction between the actuator portion **125** and the base portion **120** and weld the actuator portion **125** with the base portion **120** via friction welding. For example, the member actuator **135** can move the grip member **130** to the first position (e.g., open or stored position) such that the welding component **110** can move into the body **105**. The member actuator **135** can move the grip member **130** to the second position (e.g., a motor starting position) to facilitate actuation of a welding cycle. For example, having the grip member **130** in the second position can create a desired amount of friction between the base portion **120** and the actuator portion **125** at a beginning of a welding cycle, but not too much friction that prevents the weld actuator **140** from being able to move the actuator portion **125** relative to the base portion **120**. The weld actuator **140** can move the actuator portion **125** between the grip member **130** and the base member **115** to generate the friction needed to weld the actuator portion **125** with the base portion **120**. The member actuator **135** can be or include any type of motor. For example, the member actuator **135** can be a servo motor. The weld actuator **140** can be or include any type of motor. For example, the weld actuator **140** can be a brushless DC motor.

(26) The weld actuator **140** can engage the base member **115** to move the base member **115** relative to the welding component **110**. For example, the weld actuator **140** can move the base member **115** side to side to create friction between the actuator portion **125** and the base portion **120** and weld the actuator portion **125** with the base portion **120** via friction welding.

(27) The sensor **145** of the strapping device **100** can detect a characteristic of at least one of the components of the strapping device **100**. For example, the sensor **145** can detect a characteristic of the member actuator **135**. The characteristic can be, for example, an electrical characteristic of the member actuator **135**. The electrical characteristic can be, for example, current, power consumption, or resistance felt by the member actuator **135**. The sensor **145** can be electrically coupled with the member actuator **135** to detect the electrical characteristic of the member actuator **135**. The characteristic can be, for example, a physical characteristic of the grip member **130**. The physical characteristic can be, for example, a force felt by a surface of the grip member **130** that contacts the welding component **110** or a distance detected between the grip member **130** and the welding component **110**. The sensor **145** can be a part of the grip member **130** or be disposed in the body **105** at a location such that the sensor **145** can observe the grip member **130** (e.g., see the distance between the grip member and the welding component **110**). The sensor **145** can be any sensor capable of detecting such characteristics. For example, the sensor **145** can be a pressure sensor, a current sensor, or a distance sensor, among others.

(28) The sensor **145** can transmit a signal indicative of a detected characteristic. For example, the sensor **145** can transmit a signal to the data processing system **150** that indicates the detected characteristic.

(29) The data processing system **150** of the strapping device **100** can be disposed in the body **105** of the strapping device **100** or can be disposed outside of the strapping device (e.g., at a remote location). The data processing system **150** can receive or transmit signals between at least one of the components of the strapping device **100**. For example, the data processing system **150** can be communicably coupled with the sensor **145**. The data processing system **150** can receive a signal indicative of a characteristic of a component from the sensor **145**. For example, the data processing system **150** can receive a signal indicative of an electrical characteristic of the member actuator **135**. The data processing system **150** can store and analyze the inputs received (e.g., data, signals) received and can generate outputs (e.g., control signals) based, at least partially, on the inputs. For example, the data processing system **150** can be communicably coupled with the member actuator **135**. The data processing system **150** can generate and transmit a control signal to the member actuator **135**. The control signal can cause the member actuator **135** to move the grip member **130** to a desired position. The data processing system **150** can be communicably coupled with the weld actuator **140**. For example, the data processing system **150** can generate and transmit a control signal to the weld actuator **140**. The control signal can cause the strap actuator to initiate a welding cycle by initiating movement of the actuator portion **125** of the welding component **110**.

(30) FIG. 2B depicts an example schematic diagram of the strapping device **100**. The base member **115** can include or be coupled with the compressible member **210**. The compressible member **210** can be disposed between the base member **115** and the member actuator **135**. The member actuator **135** can be coupled with the base member **115** via the compressible member **210**. The member actuator **135** can compress or decompress the compressible member **210** to adjust a pressure that the base member **115** applies to the welding component **110**. For example, member actuator **135** can move the base member **115** to move the welding component **110** toward the grip member **130** such that the grip member **130** and the base member **115** can contact the welding component **110**. With the grip member **130** in contact with the welding component **110**, the member actuator **135** can compress the compressible member **210** to increase a pressure applied to the welding component **110** via the base member **115** or decompress the compressible member **210** to decrease the pressure applied to the welding component **110**. The base member **115** can remain stationary as the member actuator **135** adjusts the compressible member **210**.

(31) For example, the member actuator **135** can move the base member **115** between a first position and a second position to selectively engage and disengage the welding component **110** with both the grip member **130** and the base member **115**. The strapping device **100** can include a cam **220**. The member actuator **135** can be coupled with the base member **115** via the cam **220**. For example, the cam **220** can be disposed between the member actuator **135** and the base member **115**. The cam **220** can be disposed between the member actuator **135** and the compressible member **210**. The member actuator **135** can be coupled with the cam **220** and can rotate the cam **220** such that the cam **220** can move (e.g., push or pull) the base member **115** in a desired direction. With the grip member **130** in contact with the welding component **110**, the cam **220** can continue to move (e.g., compress or decompress) the compressible member **210** to adjust the pressure applied to the welding component **110** via the base member **115**.
(32) The weld actuator **140** of the strapping device **100** can couple with or interface with at least one of the welding component **110** and the grip member **130** to facilitate welding of the welding component **110**. For example, the weld actuator **140** can engage the actuator portion **125** of the welding component **110** to move the actuator portion **125** relative to the base portion **120** to create friction between the actuator portion **125** and the base portion **120** and weld the actuator portion **125** with the base portion **120** via friction welding. For example, the member actuator **135** can move the base member **115** to the first position (e.g., open or stored position) such that the welding component **110** can move into the body **105**. The member actuator **135** can move the base member **115** to the second position (e.g., a motor starting position) to facilitate actuation of a welding cycle. For example, having the base member **115** in the second position can create a desired amount of friction between the base portion **120** and the actuator portion **125** of the welding component at a beginning of a welding cycle, but not too much friction that prevents the weld actuator **140** from being able to move the actuator portion **125** relative to the base portion **120**. The weld actuator **140** can move the actuator portion **125** between the grip member **130** and the base member **115** to generate the friction needed to weld the actuator portion **125** with the base portion **120**.
(33) The weld actuator **140** can engage the grip member **130** to move the grip member **130** relative to the welding component **110**. For example, the weld actuator **140** can move the grip member **130** side to side to create friction between the actuator portion **125** and the base portion **120** and weld the actuator portion **125** with the base portion **120** via friction welding.
(34) FIG. **3** depicts side views of a plurality of example positions of the grip member **130** and the compressible member **210**. For example, the grip member **130** and the compressible member **210** can have a first position, shown as stored position **305**. The stored position **305** can be a stored or unactuated position for the grip member **130** and the compressible member **210** when the strapping device **100** is not being used. For example, the stored position **305** of the compressible member **210** can include the compressible member **210** having a first length **335**. The first length **335** can be a natural or relaxed length due to no external forces being applied to the compressible member **210**. The stored position **305** of the grip member **130** can be disposed away from the base member **115**. For example, the grip member **130** can have a member contact surface **310**. The member contact surface **310** can be configured to interface with the actuator portion **125** of the welding component **110** during a welding cycle. The member contact surface **310** can be a surface of the grip member **130** or a surface of the member grip **215**. The base member **115** can have a base contact surface **315**. The base contact surface **315** can interface with the base portion **120** of the welding component **110**. The base contact surface **315** can be a surface of the base member **115** or a surface of the base grip **205**. The stored position **305** can be a programmed or predetermined position. For example, the stored position **305** can include the member contact surface **310** being disposed a distance **320** away from the base contact surface **315**. The distance **320** can be a predetermined distance. The distance **320** can be large enough such that the welding component **110** can enter the strapping device **100** and the member contact surface **310** is not in contact with the welding component **110**.
(35) The grip member **130** and the compressible member **210** can have a second position, shown as contact position **325**. The contact position **325** can be when the member contact surface **310** of the grip member **130** initiates contact with the actuator portion **125** of the welding component **110**. The contact position **325** of the grip member **130** can be the location of the grip member **130** when the grip member **130** contacts the actuator portion **125**. For example, the member actuator **135** can move the grip member **130** in a first direction (e.g., down) from the stored position **305** until the grip member **130** contacts the actuator portion **125**. The location of the actuator portion **125** can define the contact position **325**. The contact position **325** can be a detectable position. For example, the sensor **145** can detect when the member contact surface **310** contacts the actuator portion **125** of the welding component **110**. For example, the sensor **145** can be a current sensor and detect a current that is being used by the member actuator **135** as the member actuator **135** moves the grip member **130** from the stored position **305** toward the contact position **325**. The sensor **145** can detect an increase in current when the member contact surface **310** contacts the member actuator **135** due to an increase in resistance due to the contact with the actuator portion **125**. The increase in the current can indicate the contact between the member contact surface **310** and the actuator portion **125** of the welding component **110**. The contact position **325** of the compressible member **210** can include the compressible member **210** having a second length **335**. For example, the compressible member **210** may be compressed when the grip member **130** contacts the actuator portion **125** of the welding component **110** such that the second length **335** is less than the first length **335** from the stored position **305**.
(36) The grip member **130** and the compressible member **210** can have a third position, shown as starting position (e.g., motor starting position) **330**. The starting position **330** can be the position of the grip member **130** and the compressible member **210** when a welding cycle begins. The starting position **330** can provide a desired amount of pressure to the welding component **110** such that there is enough pressure to create the friction needed to generate the weld, and not too much pressure to prevent the components of the strapping device **100** from moving to create the friction. The starting position **330** of the compressible member **210** can include the compressible member **210** having a third length **335**. For example, the member actuator **135** can adjust a pressure applied to the welding component **110** by compressing or decompressing the compressible member **210**. Compressing the compressible member **210** such that the third length **335** is less than the second length **335** can increase the pressure. Decompressing the compressible member **210** such that the third length **335** is greater than the second length **335** can

decrease the pressure. The starting position **330** of the grip member **130** can include moving the grip member **130** a distance away from the contact position **325** of the grip member **130** or keeping the grip member **130** at the contact position **325**. Moving the grip member **130** can adjust the pressure that the grip member **130** applies to the welding component **110**. The member actuator **135** can move the grip member **130** in the first direction or in a second, opposite direction (e.g., up) to move the grip member **130** from the contact position **325** to the starting position **330**. For example, the member actuator **135** can move the grip member **130** further in the first direction to increase the pressure and can move the grip member in the second, opposite direction to decrease the pressure. Adjusting the compressible member **210** can adjust the pressure enough such that the contact position **325** and the starting position **330** of the grip member **130** are the same position for the grip member **130**.

(37) The starting position **330** can be a programmable position. The starting position **330** can be a relative position. For example, the starting position **330** can be based, at least partially, on the contact position **325**. For example, the starting position **330** for the grip member **130** or the compressible member **210** can be a predetermined distance (e.g., in either direction) away from the contact position **325**. The starting position **330** can be based, at least partially, on a thickness **340** of the welding component **110**. For example, a larger thickness **340** can be associated with a larger predetermined distance and a smaller thickness **340** can be associated with a smaller predetermined distance, or vice versa. The starting position **330** can be based, at least partially, on the detected characteristic of the component of the strapping device **100**. For example, a detected current that is below a threshold may indicate that the grip member **130** or the compressible member **210** can move further in the first direction and a detected current that is above a threshold may indicate that the grip member **130** or the compressible member **210** can move in the second direction. The starting position **330** can be a motor starting position such that the weld actuator **140** actuates to initiate a welding cycle when the grip member **130** and the compressible member **210** are in the starting position **330**.

(38) The starting position **330** for at least one of the grip member **130** and the compressible member **210** can be the same as the contact position **325**. For example, the starting position **330** can include the grip member **130** remaining stationary in the contact position **325** and the member actuator **135** compressing or decompressing the compressible member **210** to adjust the pressure applied to the welding component **110**. The compressible member **210** can have a first length **335** (e.g., natural length) with the grip member **130** in the stored position **305**. The compressible member **210** can have a second length (e.g., shorter than the first length) with the grip member **130** in the contact position **325**. The compressible member **210** can have a third length (e.g., shorter or longer than the second length) when in the starting position **330** with the grip member **130** in the starting position **330**. For example, the member actuator **135** can compress the compressible member **210** from the second length **335** to the third length **335** to increase the pressure applied to the welding component **110** via the grip member **130**. The member actuator **135** can decompress (e.g., extend) the compressible member **210** from the second length **335** to the third length **335** to decrease the pressure applied to the welding component **110** via the grip member **130**. The member actuator **135** can move the grip member **130** without moving the compressible member **210** to get to the starting position **330** from the contact position **325**. The member actuator **135** can not move either of the grip member **130** and the compressible member **210** to get to the starting positing **330** from the contact position **325**. For example, with the grip member **130** and the compressible member **210** in the contact position **325**, a desired pressure is applied to the welding component **110** such that the member actuator **135** does not need to adjust either the grip member **130** or the compressible member **210** to achieve the desired pressure.

(39) The grip member **130** compressible member **210** can have a fourth position, shown as welding position **345**. The welding position **345** can include the grip member **130** and the compressible member **210** being at a position or length during at least a portion of a welding cycle or adjusting the position or length during the welding cycle to facilitate proper welding and cooling of the welding component **110**. The grip member **130** and the compressible member **210** can be in the welding position **345** as the strapping device **100** is actively welding the actuator portion **125** with the base portion **120** and as the welding component **110** cools after being welded. The welding position **345** of the compressible member **210** can include the compressible member **210** having a fourth length **335**. For example, the member actuator **135** can compress the compressible member **210** from the starting position **330** such that the fourth length **335** is less than the third length **335** to increase the pressure applied to the welding component **110**. The welding position **345** of the grip member **130** can include the grip member **130** being in a fourth position. For example, the fourth position can be closer to the base member **115** than the third position to increase the pressure applied to the welding component **110**. The welding position **345** can be a programmable position. For example, the welding position **345** can be based, at least partially, on at least one of the contact position **325** (e.g., a predetermined distance from the contact position **325**), the starting position **330** (e.g., a predetermined distance from the starting position **330**), and the thickness **340** of the welding component **110** (e.g., welding position **345** is to position the grip member **130** such that the welding component **110** is compressed to a thickness that is a predetermined percentage of the original, non-compressed thickness **340**). The welding position **345** can be a detectable position. For example, a sensor **145** can detect when the grip member **130** is pressing hard enough on the welding component **110** (e.g., the sensor **145** detects a threshold current being used by the member actuator **135** or a threshold pressure felt by the member contact surface **310**).

(40) The welding position **345** can also be the same as at least one of the contact position **325** and the starting position **330** with respect to the location of the grip member **130** and the length **355** of the compressible member **210**. The welding position **345** can be different from at least one of the contact position **325** and the starting position **330** with respect to at least one of the location of the grip member **130** and the length **335** of the compressible member **210**. For example, the welding position **345** can include the grip member **130** remaining stationary in at least one of the contact position **325** and the starting position **330** and the member actuator **135** compressing or decompressing the compressible member **210** to adjust the pressure applied to the welding component **110**. For example, the compressible member **210** can have a fourth length **335**. The fourth length **335** can be less than the third length **335** (e.g., the starting position **330**). The fourth length **335** can provide the desired amount of pressure to weld the welding component **110** together.

(41) While the above examples refer to the compressible member **210** being coupled with the grip member **130** and member actuator **135** adjusting the pressure applied to the welding component **110** by adjusting at least one of the length **335** of the compressible member **210** and the location of the grip member **130**, the same concepts can be applied with the compressible member **210** being coupled with the base member **115** and the member actuator **135** adjusting the pressure applied to the welding component **110** by adjusting at least one of the length **335** of the compressible member **210** and the location of the base member **115**. For example, the member actuator **135** can move the base member **115** toward the grip member **130**, and adjust the pressure applied to the welding component **110** by compressing or decompressing the compressible member **210** that is coupled with the base member **115**. The concepts can also be applied to a strapping device **100** that does not include a compressible member **210**. For example, the pressure applied to the welding component **110** can be adjusted solely by movement of the grip member **130** or the base member **115**.
(42) FIG. **4** depicts an example schematic diagram of the data processing system **150**. The data processing system **150** can receive, store, and analyze at least one input **405** and can generate at least one output **410**. For example, the data processing system **150** can include at least one input component **420**. The input component **420** can receive an input **405**. For example, the input **405** can be a signal from a sensor **145** or data from a user device **415**. The user device **415** can be any device capable of receiving input from a user and transmitting the input **405** to the input component **420** of the data processing system **150** (e.g., smart phone, laptop, computer, remote control, etc.). The input **405** can include any type of data or information related to the strapping device **100**, or a component thereof. For example, the input **405** may be the signal from the sensor **145** indicating a current being drawn by the strapping device **100**, a signal from the sensor **145** indicating a pressure applied to the member contact surface **310** of the grip member **130**, or information provided by a user of the user device **415** indicating a characteristic of the welding component (e.g., material, thickness, etc.) or a characteristic of the strapping device **100** (e.g., age of device, number of welding cycles performed, etc.), among others.
(43) The data processing system **150** can include at least one memory **425**. The memory **425** can store the data from the inputs **405** received by the input component **420**. For example, the memory **425** can store instructions **430** received from the user device **415**. The instructions **430** can, for example, indicate settings or positions for components of the strapping device **100**. For example, the instructions **430** can indicate a stored position **305** for the grip member **130**. The instructions **430** can indicate that the stored position **305** includes the grip member **130** being disposed a predetermined distance **320** away from the base member **115**. The predetermined distance **320** can be large enough to insert a welding component **110** into the strapping device **100**. The instructions **430** can indicate when to move certain components of the strapping device **100**, where to move the components, and how long to move the components, among others.
(44) The memory **425** can store at least one threshold **435**. The threshold **435** can be, for example, a current, an increase in current, a pressure, or a welding component thickness **340**, among others. The threshold **435** can indicate which instruction **430** to apply. For example, detection of a current by the sensor **145** that exceeds a current threshold **435** can indicate that the grip member **130** is in a contact position **325** and that the strapping device **100** can determine a starting position **330** and move the grip member **130** to the starting position **330** (if necessary). Determination that a thickness **340** of the welding component **110** exceeds a threshold **435** can indicate which instruction **430** to use to determine at least one of the starting position **330** and the welding position **345** for the grip member **130**. For example, a first thickness **340** that meets or exceeds a first threshold **435** can indicate a first starting position **330** and a second thickness **340** that meets or exceeds a second threshold can indicate a second starting position **330**.
(45) The thresholds **435** can be predefined values or can be adaptive values. For example, a threshold **435** can be adjusted or calculated in real-time (or near-real time) based on the input **405** received. For example, an electrical characteristic above a threshold **435** detected by the sensor **145** can indicate a starting position **330** for at least one component of the strapping device **100** (e.g., grip member **130**, base member **115**, compressible member **210**). However, another sensor **145** can detect that the weld actuator **140** is uses too much power (e.g., an amount of power over a threshold value) to initiate a weld cycle with the component in the starting position **330** (e.g., the starting position **330** is applying too much pressure on the welding component **110**). Accordingly, the data processing system **150** can be configured to modify the threshold **435** by, for example, adjusting the value of the threshold **435** or associating the threshold **435** with a different starting position **330**. The threshold **435** can be modified based on a user input **405** from a user device **415**. For example, a user can provide a value for a threshold **435** or an action associated with a threshold **435** from a remote location via the user device **415**. The data processing system can receive the user input **405** and apply the updated threshold **435** to future weld cycles.
(46) The data processing system **150** can include at least one processor **440**. The processor **440** can analyze the input **405** received via the input component **420** and generate an output **410**. The output **410** can be based, at least partially, on at least one of the input **405** received via the input component **420** and the data stored in the memory **425**. The processor **440** can compare the input **405** to a threshold **435**. For example, the input **405** can be a signal from the sensor **145** indicating a current being used by the member actuator **135**. The processor **440** can compare the received current to a current threshold **435**. The processor **440** can determine whether the received current meets or exceeds the current threshold **435**. The processor **440** can apply the instructions **430** to the input **405** received and to the determinations made by the processor **440**. For example, the instructions **430** can indicate that an input **405** that exceeds a threshold **435** can cause the processor **440** to generate a certain output **410**. For example, the instructions **430** can indicate that when a current meets or exceeds a current threshold **435**, the processor **440** can generate a specific control signal. The control signal can cause a component of the strapping device **100** to actuate or move. For example, the control signal can cause at least one of the member actuator **134** and the weld actuator **140** to actuate.
(47) The data processing system **150** can determine and adjust a welding cycle of the strapping device **100** based, at least partially, on a thickness **340** of a welding component **110**. For example, the data processing system **150** can determine when the grip member **130** is in contact with the welding component **110**. The data processing system **150** can determine the grip

member **130** is in contact with the welding component **110** based, at least partially, on input **405** received from the sensor **145**. For example, the input **405** can be a signal from the sensor **145** indicating an electrical characteristic. The electrical characteristic can be associated with the member actuator **135**, for example. The electrical characteristic can be a current drawn or used by the member actuator **135** to move the grip member **130** from the stored position **305** toward the welding component **110** or a pressure applied to the member contact surface **310** of the grip member **130**, among others. The data processing system **150** can compare the electrical characteristic to a threshold **435** (e.g., a predetermined electrical threshold). The threshold **435** can be based, at least partially, on a characteristic of the strapping device **100** such that the threshold **435** is tailored to the specific strapping device **100**. For example, the threshold **435** can be based on at least one of an age of the member actuator **135**, performance data of the member actuator **135**, and average performance data of a plurality of member actuators **135**. The data processing system **150** can, for example, identify an increase in the current drawn by the member actuator **135** based on the comparison of the electrical characteristic and the threshold **435** (e.g., a first signal can indicate a current below the threshold and a second signal can indicate a current above the threshold). The threshold **435** can indicate when the grip member **130** contacts the welding component **110**. For example, the increase in current can indicate an increase in mechanical resistance due to the grip member **130** being in contact with the welding component **110** and the extra power needed to push against the welding component **110**. When the current detected by the sensor **145** exceeds a threshold **435**, the data processing system **150** can determine that the grip member **130** is in contact with the welding component **110**. The threshold **435** can differentiate a change in an electrical characteristic due to the grip member **130** contacting the welding component **110** from a change in the electrical characteristic due to a different factor (e.g., older components can use more current than new components).

(48) The data processing system **150** can identify a position of the grip member **130** when the electrical characteristic meets or exceeds the threshold **435** and designate the position as the contact position **325**. For example, the data processing system **150** can identify the contact position **325** when the current drawn by the member actuator **135** increases and meets or exceeds a predetermined current threshold **435**. The data processing system **150** can store the contact position **325** in the memory **425**.

(49) The data processing system **150** can determine a thickness **340** of the welding component **110**. For example, the data processing system **150** can determine the thickness **340** based, at least partially, on the contact position **325**. For example, the data processing system **150** can determine the thickness **340** of the welding component **110** based, at least partially, on an electrical characteristic of a component of the strapping device **100** (e.g., when the electrical characteristic meets or exceeds a threshold **435**) and a distance the grip member **130** has moved from the stored position **305** when the grip member **130** first contacts the welding component **110** (e.g., when the grip member **130** is in the contact position **325**). For example, the data processing system **150** can determine a distance **320** between the base contact surface **315** and the member contact surface **310** when the grip member **130** is in the contact position **325**. The distance **320** can be the thickness **340** of the welding component **110**. A sensor **145** can, for example, detect the distance **320** between the base contact surface **315** and the member contact surface **310** or the data processing system **150** can subtract a distance that the grip member **130** has moved from an initial distance between the base contact surface **315** and the member contact surface when the grip member **130** was in the stored position **305**.

(50) The data processing system **150** can determine the starting position **330**. The starting position **330** can be based, at least partially, on at least one of the contact position **325**, the thickness **340** of the welding component **110**, and the detected characteristic. For example, the instructions **430** can indicate that the starting position **330** is a position that is relative to the contact position **325**. For example, the starting position **330** can include the grip member **130** being be a predetermined distance away from a position of the grip member **130** in the contact position **325**. The instructions **430** can indicate that the starting position **330** is based on the thickness **340**. For example, a first welding component **110** with a first thickness **340** can have a first starting position **330** with the grip member **130** a first distance away from the contact position **325** and a second welding component **110** with a second thickness **340** can have a second starting position **330** with the grip member **130** a second distance away from the contact position **325**. The instructions **430** can indicate that the starting position **330** is based on the detected characteristic. For example, the starting position **330** of the grip member **130** can be above the contact position **325** of the grip member **130** based on the detected characteristic being above a threshold **435** and the distance to the starting position **330** can be based on the amount over the threshold **435** the characteristic is. The starting position **330** of the grip member **130** can be below the contact position **325** of the grip member **130** based on the detected characteristic being below the threshold **435** and the distance to the starting position **330** can be based on the amount over the threshold **435** the characteristic is. The starting position **330** can be at the contact position **325** based on the detected characteristic being at the threshold **435**.

(51) The starting position **330** can include the compressible member **210** being compressed or decompressed from the contact position **325**. For example, the instructions **430** can indicate that the starting position **330** is a length **335** relative to the length **335** from the contact position **325**. The starting position **330** can be based, at least partially, on a pressure applied to the welding component **110**. For example, the instructions **430** can indicate a starting pressure. The member actuator **135** can adjust the length **335** of the compressible member **210** to achieve the starting pressure. The starting pressure can be based, at least partially, on the thickness **340** of the welding component **110**.

(52) The data processing system **150** can determine the welding position **345**. The welding position **345** can be based, at least partially, on the thickness **340** of the welding component **110**. For example, the instructions **430** can indicate a specific welding position **345** for a specific thickness **340**. For example, the instructions **430** for a first welding position **345** for a first welding component **110** with a first thickness **340** can have the grip member **130** at a first position or the compressible member **210** at a first length **335** and a second welding position **345** for a second welding component **110** with a second thickness **340** can have the grip member **130** at a second position or the compressible member **210** at a second length **335**. The instructions **430** can define the welding position **345** with an absolute value (e.g., a predetermined distance from the starting position **330**), a

relative value (e.g., a distance from the starting position **330** that is a predetermined percentage of the thickness **340**), or a dependent value (e.g., a distance based on when a sensor **145** detects a characteristic that meets or exceeds a threshold **435**).

(53) The data processing system **150** can determine at least one welding cycle parameter. The welding cycle parameter can include at least one of a motor starting position, a welding time, a cooling time, a holding position, a welding pressure, and an actuator speed (e.g., speed of the weld actuator **140** to move the actuator portion **125**, speed of the member actuator **135** to move the grip member **130**). The welding cycle parameter can be based, at least partially, on a characteristic of the welding component **110**. For example, the characteristic of the welding component **110** can be the thickness **340** of the welding component **110** or a material of the welding component **110**. The data processing system **150** can determine the material of the welding component **110** from an input **405** received from either a sensor **145** (e.g., sensor **145** detect activation of a switch on the strapping device **100** indicative of a specific material) or a user device **415** (e.g., user inputs a specific material to be welded). The welding cycle parameter(s) for a first welding component **110** made of a first material can be different than the welding cycle parameter(s) for a second welding component **110** made of a second material. For example, the first material may require a faster actuator speed and a longer cooling time. The welding parameter(s) for a first welding component **110** with a first thickness **340** can be different than the welding parameter(s) for a second welding component **110** with a second thickness **340**. The data processing system **150** can compare the characteristic(s) of the welding component **110** with the instructions **430** and the thresholds **435** and determine a welding cycle that is customized for the specific welding component **110** and the strapping device **100**.

(54) The data processing system **150** can generate at least one output **410**. The output **410** can be based on the input **405** received by the input component **420** and the data stored in the memory **425**. The output **410** can be a control signal that the data processing system **150** can transmit to a component of the strapping device **100** to facilitate activation and completion of a welding cycle. For example, the data processing system **150** can generate a control signal to adjust the pressure applied to the welding component **110** by either adjusting the location of the grip member **130** or compressing or decompressing the compressible member **210**. The data processing system **150** can generate a control signal to actuate the member actuator **135** to move the grip member **130** or the compressible member **210** from the contact position **325** to the starting position **330**. The data processing system **150** can generate and transmit the signal based on the grip member **130** being in contact with the welding component **110**. The data processing system **150** can generate a control signal to actuate the weld actuator **140** to move at least one of the actuator portion **125** of the welding component **110** relative to the base portion **120** and the base member **115** relative to actuator portion **125** to initiate a welding cycle to weld the welding component **110**. The data processing system **150** can transmit the control signal such that the welding cycle initiates with the grip member **130** and the compressible member **210** at the starting position **330**. The data processing system **150** can generate a control signal to actuate the member actuator **135** to move the grip member **130** or the compressible member **210** from the starting position **330** to the welding position **345**. The control signal can indicate a rate at which to move the grip member **130** or the compressible member **210** from the starting position **330** to the welding position **345**. The control signal can indicate to hold the grip member **130** or the compressible member in the welding position **345** until the end of the welding cycle. The data processing system **150** can generate a control signal to actuate the member actuator **135** to move the grip member **130** or the compressible member from the welding position **345** to the stored position **305**. The data processing system **150** can generate a control signal that indicates a welding cycle parameter (e.g., welding time, cooling time, actuator speed, etc.). The welding cycle parameter can be based on the thickness **340** of the welding component **110**. The data processing system **150** can generate any number of control signals, and each control signal can include any number of commands (e.g., a single control signal can cause multiple components to move). The data processing system **150** can generate control signals to move the components of the strapping device **100** between any of the positions in any order.

(55) The data processing system **150** can analyze the inputs **405** and generate the outputs **410** in real time or in near-real time such that the welding cycle can be a continuous process that is completed in a short amount of time. For example, the welding cycle can be completed in less than five seconds (e.g., the welding cycle can be completed in less than 3 seconds).

(56) FIG. **5** is a flow diagram depicting a method **500**. The method **500** can be a method of adjusting a strapping device **100** to accommodate different welding components **110**. The method **500** can include receiving at least one signal (Act **505**), determining a contact (Act **510**), and transmitting a control signal (Act **515**). Act **505** of receiving a signal can include a data processing system **150** receiving a signal from a sensor **145**. The signal can include data that indicates a characteristic of a component(s) of the strapping device **100**. For example, the characteristic can be an electrical characteristic. The electrical characteristics can be associated with a member actuator **135** of the strapping device **100**. The electrical characteristic can be, for example, a current or a change in the current being used by the member actuator **135** to move a member of the strapping device **100** (e.g., grip member **130**, base member **115**). Other electrical characteristics can include, for example, a resistance felt by the member actuator **135** or a pressure felt by the grip member **130** or base member **115**. The characteristic can be a physical characteristic of the strapping device **100**. For example, the physical characteristic can be a distance between two components (e.g., distance between a member contact surface **310** of the grip member **130** and an actuator portion **125** of a welding component **110**). The sensor **145** can detect the characteristic of the component(s) of the strapping device **100** and transmit a signal to the data processing system **150** indicative of the characteristic.

(57) Act **510** of determining a contact can include analyzing the received signal. For example, the data processing system **150** can compare the characteristic from the signal with a threshold **435** stored in the memory **425** of the data processing system **150**. The threshold **534** can indicate when the grip member **130** contacts the welding component **110**. For example, the signal can include data indicative of a current being used by the member actuator **135** to move the grip member **130** or the base member **115** from the stored position **305** toward the contact position **325**. The data processing system **150** can compare the current to a predetermined current threshold **435**. The threshold **435** can be a value of a current that indicates the grip member **130** is in contact with the welding component **110**. For example, the current used by the member actuator **135** can spike or

increase when the grip member **130** contacts the welding component **110** due to the force applied to the grip member **130** by the welding component **110**. The data processing system **150** can determine that a contact is made between the grip member **130** and the welding component **110** when the detected current meets or exceeds the predetermined current threshold **435**.

(58) Determining the contact at act **510** can include determining a contact position **325**. The data processing system **150** can determine the contact position **325** by measuring a distance between the member contact surface **310** and the base contact surface **315** or identifying a distance the grip member **130** moved between the stored position **305** and the contact position **325**, for example. The data processing system **150** can store the contact position in the memory **425**.

(59) Act **515** of transmitting a control signal can include generating a control signal to move at least one of the grip member **130** and the compressible member **210** from the contact position **325** to a starting position **330**. Act **515** can include the data processing system **150** determining the starting position **330**. The starting position **330** can be based, at least partially, on the contact position **325**. For example, the starting position **330** can be a predetermined distance away from the contact position **325**. The starting position **330** can be based, at least partially, on a thickness **340** of the welding component **110**. Act **515** can include the data processing system **150** determining the thickness **340** of the welding component **110**. For example, the data processing system **150** can measure or calculate the thickness **340** based on, for example, the contact position **325**. The distance between the member contact surface **310** and the base contact surface **315** when the grip member **130** is in the contact position **325** can be the thickness **340** of the welding component **110**. The instructions can associate a larger thickness **340** with a larger predetermined distance and a smaller thickness **340** with a smaller predetermined distance, or vice versa. The data processing system **150** can apply the instructions based on the determined thickness **340**. The starting position **330** can be based, at least partially, on the characteristic indicated by the signal. For example, the data processing system **150** can determine how much a current of the member actuator **135** is over or under a threshold **435**, and move the grip member **130** in a direction a distance from the contact position **325** based on the amount over or under the current it. The data processing system **150** can determine the starting position **330** is the same position as the contact position **325**.

(60) The data processing system **150** can generate the control signal based on the starting position **330**. For example, the data processing system **150** can transmit the control signal to the member actuator **135** to move at least one of the grip member **130** and the compressible member **210** to the starting position **330**. The starting position **330** can be the same as the contact position **325**. As such, the data processing system **150** can not generate a control signal to move the grip member **130** and the compressible member **210** from the contact position **325**.

(61) Act **515** of transmitting a control signal can include generating a control signal to initiate a welding cycle. Act **515** can include the data processing system **150** determining the welding cycle parameters for a welding cycle. For example, the data processing system **150** can determine at least one of a welding time, a cooling time, a member actuator **135** speed, and a weld actuator **140** speed, among others. The welding cycle parameters can be based, at least partially, on characteristics of the welding component **110** (e.g., thickness **340** and material). The data processing system **150** can transmit a signal to the member actuator **135** and the weld actuator **140** to initiate the welding cycle according to the welding cycle parameters.

(62) As an example of the method **500**, a sensor **145** can detect a current of the member actuator **135**. The sensor **145** can transmit a signal to the data processing system **150** indicating the current of the member actuator **135**. The data processing system **150** can compare the current to a predetermined current threshold **435**. The data processing system **150** can determine the current meets or exceeds the predetermined current threshold **435** and identify the contact position **325** based on the current meeting or exceeding the predetermined current threshold **435**. The data processing system **150** can determine a thickness of the welding component **110** based on the contact position **325**. The data processing system **150** can determine welding cycle parameters based on the thickness of the welding component **110**, including the starting position **330**. The starting position **330** may include the grip member **130** remaining in the same position and adjusting the length **335** of the compressible member **210** to adjust the pressure applied to the welding component **110**. The data processing system **150** can transmit a first control signal to the member actuator **135** to compress or decompress the compressible member **210** to the starting position **330** if the starting position **330** is not the same as the contact position **325**. The data processing system **150** can transmit a second control signal to the member actuator **135** and the weld actuator **140** to initiate the welding cycle in accordance with the welding cycle parameters. The second control signal can cause at least one of the grip member **130** and the compressible member **210** to move to the welding position **345** and remain in the welding position **345** until the end of the welding portion and the cooling portion of the welding cycle. After the cooling portion, the data processing system **150** can transmit a third control signal to cause the grip member **130** and the compressible member **210** to move to the stored position **305**.

(63) FIG. **6** is a flow diagram depicting a method **600**. The method **600** can be a method of adjusting a strapping device **100** to accommodate different welding components **110**. The method **600** can include detecting a contact (Act **605**), adjusting a pressure (Act **610**), and initiating welding of a welding component (Act **615**). Act **605** of detecting a contact can include moving a grip member **130** from a stored position **305** toward a welding component **110**. A sensor **145** can detect a characteristic (e.g., an electrical characteristic) of a component of a strapping device **100** as the grip member **130** moves toward the welding component **110**. For example, the sensor **145** can detect a current being used by the member actuator **135**. Act **605** can include the sensor **145** transmitting a signal to the data processing system **150** indicating the characteristic. For example, the signal can indicate the current of the member actuator **135**. Act **605** can include comparing the characteristic to a threshold **435**. The data processing system **150** can determine whether the grip member **130** has contacted the welding component **110** based on the comparison of the characteristic with the threshold **435**. For example, the data processing system **150** can determine the grip member **130** is contacting the welding component **110** when the detected current exceeds a predetermined current threshold **435**. The data processing system **150** can identify the contact position **325** of the grip member **130** based on where the grip member is when the sensor **145** detects the characteristic that meets or exceeds the threshold **435**.

(64) Act **610** of adjusting a pressure can include moving at least one of the grip member **130** and the compressible member **210**

to a starting position **330** from the contact position **325**. The data processing system **150** can determine the starting position **330** based, at least partially, on the thickness **340** of the welding component **110**. The data processing system **150** can determine at least one welding cycle parameter (e.g., welding time, cooling time, welding position, welding pressure, actuator speed, etc.) based, at least partially, on the thickness **340** of the welding component **110**. The data processing system **150** can generate a first control signal to cause the member actuator **135** to move the grip member **130** to the starting position **330** if the starting position **330** is not the same as the contact position **325**. The member actuator **135** can receive the first control signal and move the grip member **130** according to the first control signal.

(65) Act **615** of initiating welding can include initiating a welding cycle. With the grip member **130** and the compressible member **210** in the starting position **330**, the data processing system **150** can transmit a second control signal to cause the weld actuator **140** to move at least one of the actuator portion **125** of the welding component **110** and the base member according to the determined welding cycle parameters and to cause the member actuator **135** to move at least one of the grip member **130** and the compressible member **210** to the welding position **345** according to the determined welding cycle parameters.

(66) Method **600** can also apply to a strapping device **100** that includes the member actuator **135** coupled with the base member **115** and the weld actuator **140** coupled with at least one of the actuator portion **125** of the welding component **110** and the grip member **130**.

(67) Although an example computing system has been described in FIG. **4**, the subject matter including the operations described in this specification can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(68) Some of the description herein emphasizes the structural independence of the aspects of the system components or groupings of operations and responsibilities of these system components. Other groupings that execute similar overall operations are within the scope of the present application. Modules can be implemented in hardware or as computer instructions on a non-transient computer readable storage medium, and modules can be distributed across various hardware or computer based components.

(69) The systems described above can provide multiple ones of any or each of those components and these components can be provided on either a standalone system or on multiple instantiation in a distributed system. In addition, the systems and methods described above can be provided as one or more computer-readable programs or executable instructions embodied on or in one or more articles of manufacture. The article of manufacture can be cloud storage, a hard disk, a CD-ROM, a flash memory card, a PROM, a RAM, a ROM, or a magnetic tape. In general, the computer-readable programs can be implemented in any programming language, such as LISP, PERL, C, C++, C #, PROLOG, or in any byte code language such as JAVA. The software programs or executable instructions can be stored on or in one or more articles of manufacture as object code.

(70) Example and non-limiting module implementation elements include sensors providing any value determined herein, sensors providing any value that is a precursor to a value determined herein, datalink or network hardware including communication chips, oscillating crystals, communication links, cables, twisted pair wiring, coaxial wiring, shielded wiring, transmitters, receivers, or transceivers, logic circuits, hard-wired logic circuits, reconfigurable logic circuits in a particular non-transient state configured according to the module specification, any actuator including at least an electrical, hydraulic, or pneumatic actuator, a solenoid, an op-amp, analog control elements (springs, filters, integrators, adders, dividers, gain elements), or digital control elements.

(71) The subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. The subject matter described in this specification can be implemented as one or more computer programs, e.g., one or more circuits of computer program instructions, encoded on one or more computer storage media for execution by, or to control the operation of, data processing apparatuses. Alternatively or in addition, the program instructions can be encoded on an artificially generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. While a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate components or media (e.g., multiple CDs, disks, or other storage devices include cloud storage). The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

(72) The terms “computing device”, “component” or “data processing system” or the like encompass various apparatuses, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations of the foregoing. The apparatus can include special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(73) A computer program (also known as a program, software, software application, app, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and can be

deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program can correspond to a file in a file system. A computer program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.
(74) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatuses can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Devices suitable for storing computer program instructions and data can include non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.
(75) The subject matter described herein can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a web browser through which a user can interact with an implementation of the subject matter described in this specification, or a combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).
(76) While operations are depicted in the drawings in a particular order, such operations are not required to be performed in the particular order shown or in sequential order, and all illustrated operations are not required to be performed. Actions described herein can be performed in a different order.
(77) Having now described some illustrative implementations, it is apparent that the foregoing is illustrative and not limiting, having been presented by way of example. In particular, although many of the examples presented herein involve specific combinations of method acts or system elements, those acts and those elements may be combined in other ways to accomplish the same objectives. Acts, elements and features discussed in connection with one implementation are not intended to be excluded from a similar role in other implementations or implementations.
(78) The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of "including" "comprising" "having" "containing" "involving" "characterized by" "characterized in that" and variations thereof herein, is meant to encompass the items listed thereafter, equivalents thereof, and additional items, as well as alternate implementations consisting of the items listed thereafter exclusively. In one implementation, the systems and methods described herein consist of one, each combination of more than one, or all of the described elements, acts, or components.
(79) Any references to implementations or elements or acts of the systems and methods herein referred to in the singular may also embrace implementations including a plurality of these elements, and any references in plural to any implementation or element or act herein may also embrace implementations including only a single element. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements to single or plural configurations. References to any act or element being based on any information, act or element may include implementations where the act or element is based at least in part on any information, act, or element.
(80) Any implementation disclosed herein may be combined with any other implementation or embodiment, and references to "an implementation," "some implementations," "one implementation" or the like are not necessarily mutually exclusive and are intended to indicate that a particular feature, structure, or characteristic described in connection with the implementation may be included in at least one implementation or embodiment. Such terms as used herein are not necessarily all referring to the same implementation. Any implementation may be combined with any other implementation, inclusively or exclusively, in any manner consistent with the aspects and implementations disclosed herein.
(81) References to "or" may be construed as inclusive so that any terms described using "or" may indicate any of a single, more than one, and all of the described terms. References to at least one of a conjunctive list of terms may be construed as an inclusive OR to indicate any of a single, more than one, and all of the described terms. For example, a reference to "at least one of 'A' and 'B'" can include only 'A', only 'B', as well as both 'A' and 'B'. Such references used in conjunction with "comprising" or other open terminology can include additional items.
(82) Where technical features in the drawings, detailed description or any claim are followed by reference signs, the reference signs have been included to increase the intelligibility of the drawings, detailed description, and claims. Accordingly, neither the reference signs nor their absence have any limiting effect on the scope of any claim elements.
(83) Modifications of described elements and acts such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations can occur without materially departing from the teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed can be constructed of multiple parts or elements, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Other substitutions, modifications, changes and omissions can also be made in the design, operating conditions and arrangement of the disclosed elements and operations without departing from the scope of the present disclosure.
(84) For example, descriptions of up and down may be reversed. Further relative parallel, perpendicular, vertical or other positioning or orientation descriptions include variations within +/−10% or +/−10 degrees of pure vertical, parallel or

perpendicular positioning. References to “approximately,” “substantially” or other terms of degree include variations of +/−10% from the given measurement, unit, or range unless explicitly indicated otherwise. Coupled elements can be electrically, mechanically, or physically coupled with one another directly or with intervening elements. Scope of the systems and methods described herein is thus indicated by the appended claims, rather than the foregoing description, and changes that come within the meaning and range of equivalency of the claims are embraced therein.

Claims

1. A strapping device, comprising: a first member; a second member opposite the first member; a first actuator coupled with the first member to move the first member between a first position and a second position; a sensor to detect an electrical characteristic of the first actuator with the first member at the first position and to transmit a signal indicative of the electrical characteristic; and a data processing system configured to: receive the signal indicative of the electrical characteristic; determine the first member is in contact with a welding component based on the electrical characteristic; determine the second position of the first member based on the first member being in contact with the welding component; transmit a first control signal to the first actuator to move the first member to the second position to adjust a pressure the first member applies to the welding component; and transmit a second control signal to the first actuator and a second actuator to initiate a welding cycle to weld the welding component such that the welding cycle initiates with the first member at the second position.
2. The strapping device of claim 1, comprising: the data processing system configured to: compare the electrical characteristic with a predetermined electrical threshold, the predetermined electrical threshold based on at least one of an age of the first actuator, performance data of the first actuator, and average performance data of other actuators.
3. The strapping device of claim 1, comprising: the data processing system is configured to: identify an increase in a current of the first actuator based on the electrical characteristic; identify a contact position of the first member when the increase in the current occurs; determine a thickness of the welding component based on the contact position; and determine the second position based on the thickness of the welding component.
4. The strapping device of claim 1, comprising: the data processing system is configured to: determine a thickness of the welding component based on the electrical characteristic; determine a distance the first member has moved from a stored position when the first member contacts the welding component at the first position; and determine the second position based on the thickness of the welding component.
5. The strapping device of claim 1, comprising: the data processing system is configured to determine a thickness of the welding component; and the control signal comprising a welding cycle parameter based on the thickness of the welding component, the welding cycle parameter comprising at least one of a welding time, a cooling time, a welding pressure, and an actuator speed for at least one of the first actuator and the second actuator.
6. The strapping device of claim 1, comprising: the electrical characteristic being a current used to power the first actuator, an increase in the current indicative of an increase of mechanical resistance due to the first member being in contact with the welding component.
7. The strapping device of claim 1, comprising: the first actuator to move the first member in a first direction to contact the welding component; and the first actuator to move the first member in a second direction to move the member to the second position, the second direction opposite the first direction.
8. The strapping device of claim 1, comprising: the first member to interface with a first portion of the welding component; the second member to interface with a second portion of the welding component; and the second actuator to move the first portion of the welding component relative to the second portion to weld the first portion with the second portion via friction welding.
9. The strapping device of claim 1, comprising: the second member to support a first portion and a second portion of the welding component; a first grip coupled with the first member, the first grip to interface with the first portion of the welding component and increase friction between the first portion of the welding component and the first member during the welding cycle; and a second grip coupled with the second member, the second grip to prevent the second portion of the welding component from moving; the first portion and the second portion of the welding component disposed between the first grip and the second grip.
10. The strapping device of claim 1, comprising: the second position is a motor starting position such that the second actuator actuates to initiate the welding cycle with the first member in the motor starting position.
11. The strapping device of claim 1, comprising: the data processing system is configured to: determine a thickness of the welding component; and determine a third position for the first member, the third position based on the thickness of the welding component; the control signal to cause the first actuator to move the first member to the third position during the welding cycle and hold the first member at the third position during cooling of the welding component.
12. The strapping device of claim 1, comprising: a cam coupled with the first actuator, the first actuator to rotate the cam and the cam to move the first member.
13. The strapping device of claim 1, comprising: a compressible member disposed between the first member and the first actuator, the first actuator to compress or decompress the compressible member to adjust a pressure applied to the welding component via the first member.
14. The strapping device of claim 1, comprising: the second position of the first member to provide a desired pressure to the welding component, the desired pressure based on a thickness of the welding component.
15. A method, comprising: moving a first member of a strapping device toward a second member of the strapping device via an actuator of the strapping device to a first position, a welding component disposed between the first member and the second member; receiving a signal indicative of an electrical characteristic of the actuator; determining at least one of the first member

and the second member is contacting the welding component based on the electrical characteristic; determining a second position for the first member based on the at least one of the first member and the second member contacting the welding component; transmitting a first control signal to the actuator to move the first member to the second position to adjust a pressure the first member applies to the welding component; and transmitting a second control signal to initiate a welding cycle to friction weld the welding component such that the welding cycle initiates with the first member at the second position.

16. The method of claim 15, comprising: detecting the electrical characteristic of the actuator, the electrical characteristic indicative of the one of the first member and the second member contacting the welding component; and comparing the electrical characteristic with an electrical threshold, the electrical threshold based on at least one of an age of the actuator, performance data of the actuator, and average performance data of a plurality of other actuators.

17. The method of claim 15, comprising: detecting an increase in a current of the actuator; identifying a contact position of the at least one of the first member and the second member when the increase in the current occurs; determining a thickness of the welding component based on the contact position; and determining the second position based on the thickness of the welding component.

18. The method of claim 15, comprising: determining a thickness of the welding component; and adjusting a welding cycle parameter based on the thickness of the welding component, the welding cycle parameter comprising at least one of a welding time, a cooling time, a welding pressure, and an actuator speed for the welding cycle.